THE INTERSECTION PRODUCT WITH THE FUNDAMENTAL CLASS OF AN EQUIDIMENSIONAL REGULARLY EMBEDDED SUBSCHEME IS THE GYSIN PULLBACK PUSHED FORWARD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FFE (sources/stacks-chow/pinned-original-source.tex line 14563).

Allowed standard prerequisites (editorial).

- Identification of Chow groups with operational Chow cohomology for a smooth scheme. Same chapter, Lemma labelled lemma-identify-chow-for-smooth (source line 14403), invoked at proof lines 14572-14573 and 14583. Supplies the representation alpha = c cap [X]_n and the final translation back to the intersection product.
- Commutation of the Gysin map with capping by a bivariant class. Same chapter, Lemma labelled lemma-lci-gysin-commutes (source line 13613), invoked at proof lines 14579-14580. Supplies i^!(c cap [X]_n) = c cap i^![X]_n, the second step of the chain.
- The Gysin pullback of the fundamental class along a regular closed immersion. Same chapter, Lemma labelled lemma-lci-gysin-easy (source line 13643), invoked at proof line 14581. Supplies i^![X]_d = [Y]_e in CH_*(Y); it is counted separately in this lane as S12-algebra-lci-gysin-easy.

Source reference. Original lemma 0FC2: sources/stacks-chow/pinned-original-source.tex, source lines 14403–14403. Same-chapter pointer into the author environment lemma at source line 14403; invoked at proof line 14573 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FF6: sources/stacks-chow/pinned-original-source.tex, source lines 13613–13613. Same-chapter pointer into the author environment lemma at source line 13613; invoked at proof line 14580 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FF7: sources/stacks-chow/pinned-original-source.tex, source lines 13643–13643. Same-chapter pointer into the author environment lemma

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at source line 13643; invoked at proof line 14581 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let k be a field. Let X be a scheme smooth over k. Let $i : Y \to X$ be a regular closed immersion. Let $\alpha \in \mathrm{CH}_*(X)$. If Y is equidimensional of dimension e, then $\alpha \cdot [Y]_e = i_*(i^!(\alpha))$ in $\mathrm{CH}_*(X)$.*

Proof. After decomposing X into connected components we may and do assume X is equidimensional of dimension d. Write $\alpha = c \cap [X]_n$ with $x \in A^*(X)$, see Lemma 0FC2. Then

$$i_*(i^!(\alpha)) = i_*(i^!(c \cap [X]_n)) = i_*(c \cap i^![X]_n) = i_*(c \cap [Y]_e) = c \cap i_*[Y]_e = \alpha \cdot [Y]_e$$

The first equality by choice of c. Then second equality by Lemma 0FF6. The third because $i^![X]_d = [Y]_e$ in $\mathrm{CH}_*(Y)$ (Lemma 0FF7). The fourth because bivariant classes commute with proper pushforward. The last equality by Lemma 0FC2. □